Evaluating the Washburn Equation for Biscuit Dunking Using Machine Learning Techniques: SCIFM0002 - Programming-Project

Student ID: 2724184
*Scientific Computing & Data Science, University of Bristol

Abstract

This project investigates capillary absorption in three biscuit types (Digestive, Hobnob, and Rich Tea) and compares machine learning approaches to the Washburn equation. Supervised classification models were first used to identify biscuit type from experimental absorption variables, with a Neural Network achieving 91.0% accuracy. Radii per biscuit was analysed and trajectories of time resolved data were compared to Washburn and ML predictions for predicting penetration length Machine learning improved prediction for Hobnob and Digestive biscuits, reducing RMSE by 51.2% and 43.1% respectively, while Rich Tea remained better described by the Washburn model. This suggests that simpler porous systems are well captured by classical capillary theory, whereas structurally complex biscuits deviate from ideal Washburn behaviour and benefit from data-driven modelling.

1 Introduction

Here at McVities we are constantly pushing the boundaries of biscuit manufacturing to develop new products. However, all biscuits have a vital function that they must perform to attract a British customer, that being the ability to dunk them in tea.

Biscuits are composed of a complex network of gluten fibres, sugars, fats, and air pockets, which create a porous internal structure containing microscopic channels that act as capillaries, exhibiting capillary action to draw the liquid through the biscuit, resulting in a softening in the structure. This process can be modelled using the Washburn Equation[1].

$$L = \sqrt{\frac{\gamma r t \cos\phi}{2\eta}} \tag{1}$$

Where L is the penetration distance of the liquid, γ is the surface tension of the tea, r is the capillary pore radius, t is the time of absorption, ϕ is the contact angle between the tea and biscuit surface, and η is the dynamic viscosity of the liquid. The equation predicts that the penetration distance should increase proportionally to the square root of time, $L^2 \propto t$.

However, real biscuits are markedly more complex, as their pore structures are highly irregular, with non-uniform pore sizes, branching pathways, and varying connectivity[2]. This irregular geometry means liquid flow is unlikely to follow the Washburn's assumptions. In addition, biscuits swell during absorption[3], changing the effective pore radius and deforms the biscuit leading to variations in the contact angle with the liquid. These effects may cause systematic deviations from ideal Washburn behaviour.

Machine learning(ML) models could provide a suitable alternative for modelling this system. As unlike analytical models, ML approaches are not based on strict physical assumptions and can instead learn complex nonlinear relationships directly from experimental data[4]. Comparing ML performance to the Washburn equation will allow for quantification of where the physical model succeeds and where its assumptions begin to fail, further informing biscuit design.

2 Results & Discussion

Principal Component Analysis (PCA) was applied to **dunking_data** to investigate whether the experimental variables showed clear linear separation and thus whether unsupervised clustering of biscuit types could be observed. The cumulative explained variance plot shown in Fig. 1a demonstrates that the leading components capture only a limited proportion of the total variance, with several components required before most of the variance is explained. This indicates that no single variable dominates the structure of the dataset, and instead that multiple experimental parameters contribute meaningfully to biscuit behaviour.

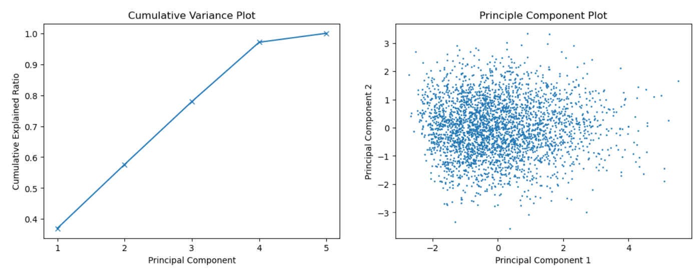

Figure 1 (a) Cumulative explained variance ratio for the principal components of the dunking dataset, showing that multiple components are required to capture most of the variance. (b) Projection of the first two principal components, showing no clear clustering of biscuit classes.

Additionally, Fig. 1b shows no clear clustering of biscuit types within the first two principal components. The dominant variance in the dataset therefore does not align strongly with biscuit class separation, and a simple linear projection is insufficient for distinguishing between biscuit classes. This suggests that low-dimensional unsupervised methods such as PCA are not suitable for classification in this system. The ineffectiveness of unsupervised learning motivates the use of supervised ML methods capable of learning more complex decision boundaries between biscuit types.

Biscuit classification was performed using the dataset `dunking_data.csv`, which contains 3000 observations with balanced class representation (1000 samples each of Rich Tea, Hobnob, and Digestive biscuits) and the five experimental parameters(γ, ϕ, η, t, L).

Several of the measured variables, particularly surface tension, viscosity, and immersion time, are controlled experimental conditions rather than intrinsic biscuit properties, as such, these influence how absorption occurs, but are not physical features distinguish biscuit type. Instead, biscuit classification emerges from how the biscuit responds to these conditions, primarily through the penetration distance L, which reflects the underlying pore structure.

Five classification models of varying complexity were tested: Logistic Regression, Support Vector Machines (SVM), Random Forest, a Custom Decision Tree, and a Neural Network. These models were chosen to compare both linear and nonlinear approaches to classification. Performance was evaluated using accuracy, precision, recall, and F1 score, as shown in Table 1.

Table 1 Comparison of classification model performance for biscuit prediction using accuracy, precision, recall, and F1 score.

Model	Accuracy	Precision	Recall	F1
Logistic Regression	0.727	0.724	0.727	0.725
SVM	0.867	0.872	0.867	0.867
Random Forest	0.832	0.829	0.832	0.830
Custom Decision Tree	0.767	0.767	0.767	0.767
Neural Network	0.910	0.910	0.910	0.910

The Neural Network achieved the highest performance across all metrics, with an accuracy of 91.0%, as such it was selected to be the final model applied to unseen data. SVM also performed strongly, achieving an accuracy of 86.7%, while Random Forest showed competitive but slightly weaker performance. The comparatively poorer performance of Logistic Regression and decision trees suggests that biscuit classes cannot be separated effectively using simple linear decision boundaries. In contrast, the stronger performance of the SVM, using an RBF kernel, and the Neural Network indicates that nonlinear relationships exist between the experimental variables and biscuit type. This supports the hypothesis that biscuit absorption behaviour is governed by the interaction between multiple parameters rather than a single dominant variable, and that flexible models are required to capture this behaviour.

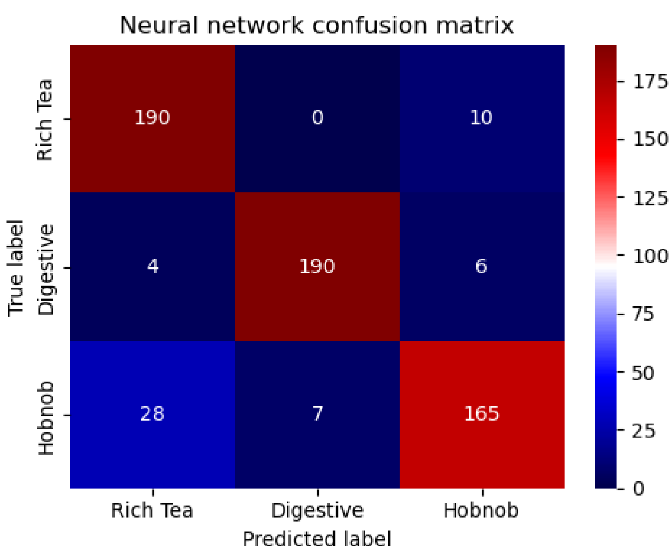

Figure 2 Confusion matrix for the Neural Network classifier showing predicted and true biscuit labels for the test set of `dunking_data.csv`.

The confusion matrix for the Neural Network classifier is shown in Fig. 2. Rich Tea and Digestive biscuits were classified most accurately, with 190 of 200 test samples correctly identified for each class. Rich Tea samples were only misclassified as Hobnobs, with no confusion with Digestives, suggesting that these two classes are well separated in feature space. Digestives also showed strong classification performance, with only minor misclassification into

Table 2 Mean pore radius, variance and number of classifications by biscuit type

Biscuit	Mean Pore Radius(m)	Variance	Counts
Digestive	7.96×10^{-7}	4.74×10^{-15}	171
Hobnob	4.96×10^{-7}	7.19×10^{-15}	169
Rich Tea	3.06×10^{-7}	4.04×10^{-15}	160

the Rich Tea and Hobnob classes. Hobnobs showed the greatest level of misclassification, with 165 of 200 samples correctly classified and the remaining samples assigned to both Rich Tea and Digestive classes. This suggests that Hobnobs occupy a more intermediate position within the feature space and share characteristics with both of the other biscuit types.

The Neural Network trained on `dunking_data` was then applied to `microscopy-data.csv`, allowing biscuit labels to be assigned and the effective pore radius required for fitting the Washburn equation (Eq. 1) to be determined.

The pore radius distributions strongly support the classification results shown in Fig. 2. Digestives exhibit the largest mean pore radius (7.96×10^{-7} m) with low variance, explaining their high classification accuracy and distinct separation in feature space. Rich Tea shows the smallest mean radius, while Hobnobs occupy an intermediate position with the largest variance, consistent with their higher misclassification rate and overlap with both other biscuit types. This suggests that pore radius is an important structural feature governing both biscuit classification and capillary absorption behaviour.

Fig. 3 also shows extended tails and overlap between the distributions. Rich Tea contains several unusually large radii, while Hobnob and Digestive overlap at intermediate values, likely reflecting misclassification caused by radius not being included in the training data. Direct regression models were also tested to predict pore radius from `dunking_data`, but these performed worse than using microscopy-derived mean radius values, so the latter was retained for Washburn modelling.

The Washburn equation was then used to predict the full time evolution of capillary rise for the unlabelled time-resolved datasets (`tr1-3`), where L was measured as a function of time under fixed experimental conditions (γ, η, ϕ). Comparing these trajectories with Washburn predictions allowed both identification of the unknown biscuit type and direct comparison between the classical physical model and machine learning methods.

Table 3 Reduced χ^2 values for each time-resolved dataset compared against the three Washburn biscuit hypotheses generated from mean radius. Lower values indicate better agreement relative to experimental uncertainty.

Dataset	Digestive	Hobnob	Rich Tea	Assigned Label
tr1	573.39	5.33	510.52	Hobnob
tr2	1571.45	374.68	11.99	Rich Tea
tr3	2495.81	17425.85	38686.33	Digestive

Table 3 shows the reduced χ^2 values obtained by comparing the Washburn-predicted penetration trajectories with the time-resolved datasets. The minimum χ^2 value allows assignment of the unknown biscuit type for each dataset. Hobnob and Rich Tea show much lower χ^2 values, indicating that the classical Washburn model provides a better fit to the experimental data, however, it is far from a strong model as indicated by χ^2. In contrast, Digestive biscuits produce substantially larger values, showing a strong

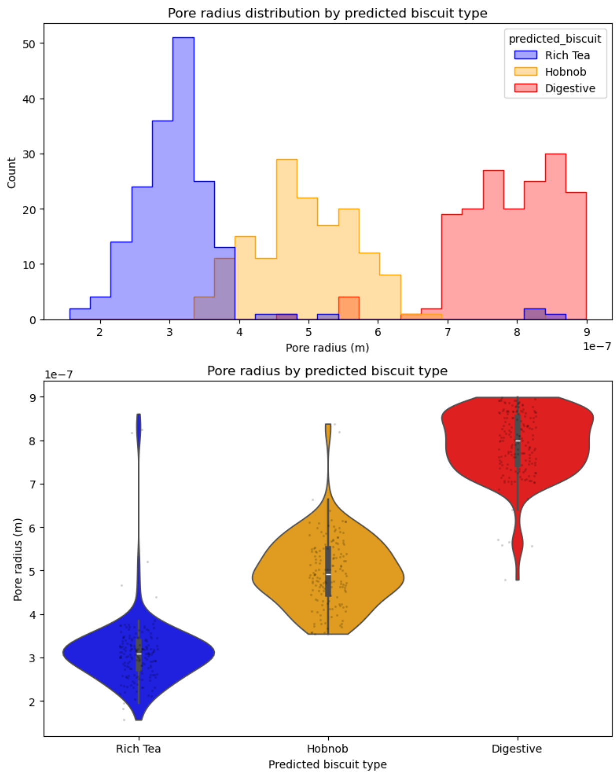

Figure 3 Summary statistics for the radii per biscuit type for microscopy_data.csv

disagreement between the physical model and the observed trajectory, motivating the use of a machine learning-based approach.

Three separate Linear Regression models were trained (one per biscuit type from `dunking_data`) to predict the penetration distance through time. The results of this can be seen in Fig.4 which demonstrates the characteristic $L^2 \propto t$ relationship which is unique per biscuit. The error was quantified in each as $d(L^2) \approx 2LdL$ owing to the squaring of L.

The results of these models can be seen in Tab.4 where systematic deviations observed between biscuit types show that the validity of the Washburn model is highly structure-dependent. Rich Tea is best described by the classical model, with lower RMSE, smaller normalised residuals, and reduced χ^2, suggesting that its more uniform pore structure satisfies the Washburn assumptions of constant capillary flow and effective pore radius.

In contrast, Digestive biscuits show the largest deviation from ideal Washburn behaviour. Although linear regression improves prediction accuracy, both the normalised residuals and reduced χ^2 remain far from the experimental uncertainty, indicating that neither model fully captures the experimental behaviour within uncertainty. This suggests that a single effective pore radius is insufficient to describe the broader and more complex pore network observed in Fig. 3.

Hobnobs occupy an intermediate position, where machine learning provides the strongest practical improvement, the Washburn model remains a valid fit. This supports the conclusion that machine learning is most useful where the assumptions of the physical model begin to fail, rather than as a complete replacement for physical theory.

3 Future Work

To address the limitations of Digestive modelling, a more advanced hybrid approach combining physics and machine learning should be developed. This is motivated by the broad pore radius distribution within the Digestive class, suggesting that the assumption of a single constant radius in the Washburn equation is insufficient, as the effective radius likely changes during absorption. Additional quantification of this could be obtained by calculating the effective pore radius as a function of time using the time-resolved datasets. If the inferred radius changes systematically with time, this would provide direct evidence that the assumption of constant pore radius is invalid and explain why the classical Washburn model fails for more structurally complex biscuits.

Following this, physics-informed machine learning models could be developed to account for time-dependent structural changes such as variation in pore radius $r(t)$ or viscosity $\eta(t)$ during absorption, allowing the model to better capture non-ideal capillary flow.

A longer-term industrial application would be the inverse design of biscuit structure using machine learning. Rather than predicting dunking behaviour from an existing biscuit, the ideal pore radius and structural properties required for desired absorption profile, such as maximising dunk time before structural failure could be formed. Similar to inverse design methods used in materials science[5] and drug discovery[6], this would allow for McVitie's to optimise biscuit formulation before manufacturing, making ML a development tool rather than limiting it to a post-production analytical method.

4 Conclusion

The results of this investigation show that the effectiveness of the Washburn equation is strongly dependent on biscuit structure. Rich Tea biscuits were best described by the classical analytical model, with lower RMSE, smaller normalised residuals, and reduced χ^2 than the machine learning regressor. This suggests that Rich Tea more closely satisfies the assumptions of uniform pore geometry and constant capillary flow, making the Washburn approximation sufficient for predicting capillary rise.

In contrast, Digestive biscuits showed the greatest deviation from ideal Washburn behaviour. Although linear regression reduced the RMSE by 43.1%, both the normalised residuals and reduced χ^2 was not a good match for experimental data, showing that neither model adequately described the data within experimental uncertainty. Hobnobs occupied an intermediate position, with machine learning providing the strongest practical improvement and predictions falling close to the experimental uncertainty.

The Neural Network classifier achieved 91.0% accuracy and allowed biscuit labels to be assigned to the microscopy dataset. The resulting pore radius distributions supported the physical interpretation of the model results, with Digestives having the largest mean radius, Rich Tea the smallest, and Hobnobs occupying an intermediate structure. However, attempts to predict pore radius directly using machine learning were less successful, so microscopy sourced mean radius remained the most robust parameter for Washburn modelling.

Overall, the Washburn equation remains effective for simpler porous systems, while machine learning becomes most valuable where the assumptions of the analytical model begin to fail. Rather than replacing physical theory, machine learning will be best applied in unison with physical models to correct for limitations.

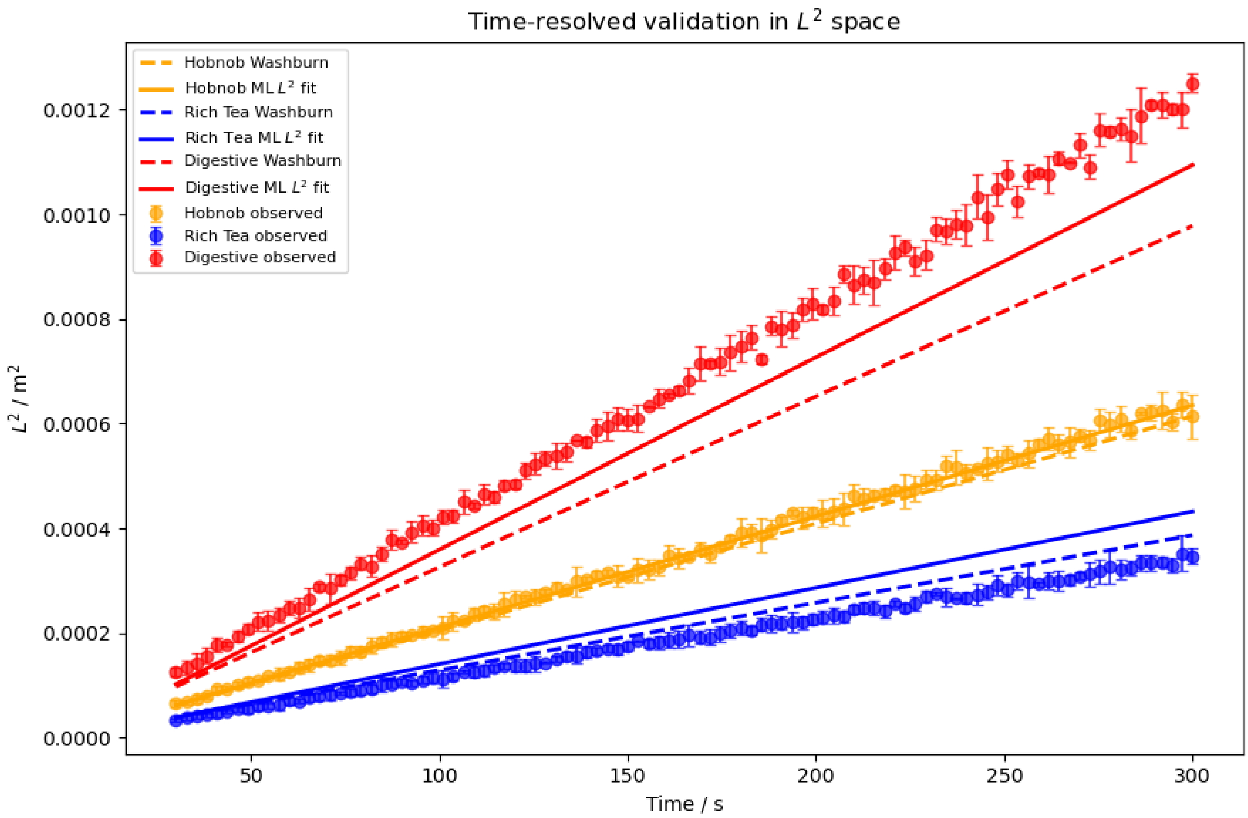

Figure 4 Time-resolved comparison of observed L^2 against time with error bars, the linear regression prediction, and the Washburn prediction calculated using experimental conditions and the mean pore radius for each assigned biscuit.

	RMSE in L		RMSE Improvement	Mean $\lvert\text{Residual}\rvert/dL$		Reduced χ^2	
Assigned Biscuit	Washburn	ML	(%)	Washburn	ML	Washburn	ML
Hobnob	0.000428	0.000209	51.2	1.436	0.676	5.33	1.24
Rich Tea	0.000828	0.001586	-91.5	2.494	4.753	11.87	43.33
Digestive	0.002916	0.001660	43.1	16.362	9.293	2470.85	817.30

Table 4 Comparison of Washburn predictions and machine learning regression for penetration length L across the three time-resolved datasets. Positive RMSE improvement indicates that the machine learning model outperformed the classical Washburn model, while negative values indicate poorer performance. The mean normalised residual and reduced χ^2 compare prediction error directly against experimental uncertainty.

References

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